

Data, Software, and Credit: Bank Lending Constraints and Digital Investment in Europe

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Background

Motivation and research question

- ▶ **Overall context:** Rising of digital technologies; need to scale up private-sector investment in this area to improve Europe's competitiveness and growth (Draghi 2024)

⇒ **Key focus:** Firms' investment in software & databases (“data-related investment”) in Europe

- ▶ This **data-related investment** is largely *intangible*, not physical capital.
- ▶ **Research question:** Do financial frictions hinder productivity through data-related investment across European firms?

⇒ Empirical assessment of productivity, investment behaviour and financing conditions at the firm level, combining **EIBIS** and **Orbis**.

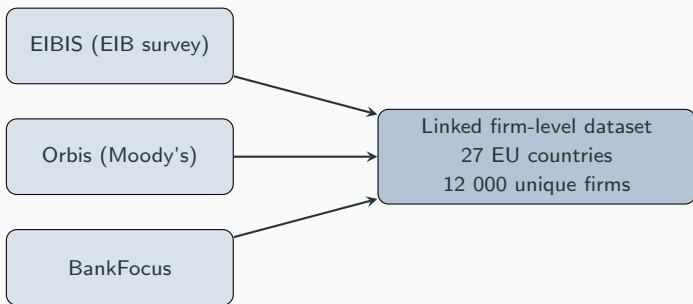
A two-step answer:

1. Firms that invest more in data-related assets have **higher labour productivity and TFPR**; this relationship is supported by IV estimates.
2. The bank credit-supply channel is much clearer for tangible investment; the data-related estimate is imprecise.

Related literature and contribution

- ▶ **AI, software, and productivity:** AI adopters tend to be more productive, but gains depend strongly on complementary digital capabilities/adoption (Babina et al. 2024; Eisefeldt et al. 2023; Calvino and Fontanelli 2023). → [Here: European firm-level evidence linking data-related investment to labour productivity and TFPR.](#)
- ▶ **Finance and firm investment:** Financing frictions reduce investment, innovation, and productivity (Ferrando and Ruggieri 2015; Duval, Hong, and Timmer 2020; Aghion et al. 2012; Hall et al. 2016). → [Here: which financing constraints bind data-related investment, relative to other assets.](#)
- ▶ **Asset heterogeneity:** Financing frictions differ across tangible and intangible assets (Rampini and Viswanathan 2013; Falato et al. 2022; Döttling and Ratnovski 2023; Demmou, Franco, and Stefanescu 2020). → [Here: constraints bind tangible investment most strongly; bank credit plays a weaker role for data-related investment.](#)

Data and Empirical Strategy



- **EIBIS (annual survey):** investment by asset type, including software & databases; AI/Big Data adoption; financing constraints; main bank.
- **Orbis:** productivity, balance sheets and firm characteristics.
- **BankFocus:** main-bank balance-sheet stress
- **Coverage:** EIBIS–Orbis sample, **2015–2025**; bank-linked sample, **2015–2020**. EIBIS covers SMEs and large firms across manufacturing, services, construction and infrastructure.

Empirical strategy

- ▶ Cross-sectional empirical strategy using survey data matched with ORBIS.
- ▶ Firm-level regressions linking:
 - ▷ data-related investment to labour productivity and TFP; and
 - ▷ bank lending constraints to investment across asset types.
- ▶ How do we attempt to identify a causal link?
 - ▷ **Productivity:** instrument data-related investment with a compute performance-per-cost measure interacted with firms' 2015 data-related investment share.
 - ▷ **Bank lending constraints:** exploit a bank credit-supply channel based on firms' exposure to their main bank's non-performing loan (NPL) ratio (update work in progress).

Specification: data-related investment and productivity

$$Prod_{i,s,c,t} = \alpha + \beta DI_{i,s,c,t} + \gamma X_{i,s,c,t} + \mu_{c \times s \times t} + \epsilon_{i,s,c,t} \quad (1)$$

Dependent var. <i>Prod</i>	Labour productivity = log(VA/employees) Revenue productivity (log TFPR ACF)
Main regressor <i>DI</i>	Data investment (software & database) AI/Big Data adoption (use dummy)
Controls <i>X</i>	Size, profitability, age, ownership
Fixed effects	Country $\times$ sector $\times$ year FE ($\mu_{c \times s \times t}$)

Note: i =firm, s =sector, c =country, t =year.

Specification: constraints and data-related investment

$$DI_{i,s,c,t} = \alpha + \beta FC_{i,s,c,t} + \gamma X_{i,s,c,t} + \mu_{c \times s \times t} + \epsilon_{i,s,c,t} \quad (2)$$

Dependent var. DI	Data investment (software & databases) AI/Big Data adoption (use dummy)
Main regressor FC	Survey-based indicators (finance, collateral, experience, investment constraints) Balance-sheet ratios: leverage, ICR
Controls X	Size, age, profitability, cashflow, ownership
Fixed effects	Country $\times$ sector $\times$ year FE ($\mu_{c \times s \times t}$)

Note: i =firm, s =sector, c =country, t =year.

Moving towards a causal link

(1) Compute performance-per-cost measure for productivity

- ▶ $Z_{i,s,c,t} = Z_t \times Z_{i,s,c}$, where Z_t is a measure of compute performance per cost and $Z_{i,s,c}$ is the firm's 2015 data-related investment share (Bergeaud 2024).
- ▶ Exploits differential exposure to improvements in the price-performance of computing based on firms' pre-existing data-related investment.

(2) Bank credit-supply channel for financing constraints

- ▶ Each firm is matched to its main bank; we assign that bank's annual NPL ratio: $Z_{it} = \text{NPLratio}_{b(i),t}$ (Brunello et al. 2022).
- ▶ Higher bank NPLs $\Rightarrow$ tighter lending standards / weaker credit supply.
- ▶ Conditional on $c \times s \times t$ FE, NPL variation captures differences in credit conditions facing the firm.

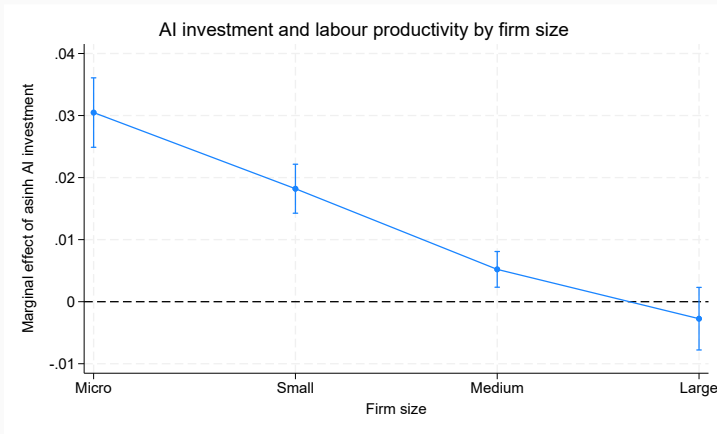
Empirical results

Outcome: log labour productivity

	OLS	OLS, IV sample	IV
<i>Investing firms only, log</i>			
Data-related inv.	0.0623*** (0.0078)	0.0527*** (0.0148)	0.0531*** (0.0174)
Observations	40,287	7,375	7,375
First-stage F	—	—	8,429.2
<i>All firms incl. zeros, asinh</i>			
Data-related inv.	0.0104*** (0.0015)	0.0102*** (0.0030)	0.0157*** (0.0050)
Observations	68,297	15,185	15,185
First-stage F	—	—	3,203.0

All specifications: firm controls, $C \times S \times Y$ FE, SE clustered $C \times S$. Instrument: compute performance-per-cost measure $\times$ 2015 data-related investment share. *** $p < 0.01$.

Productivity and firm size



→ The productivity association of data-related investment is strongest among smaller firms.

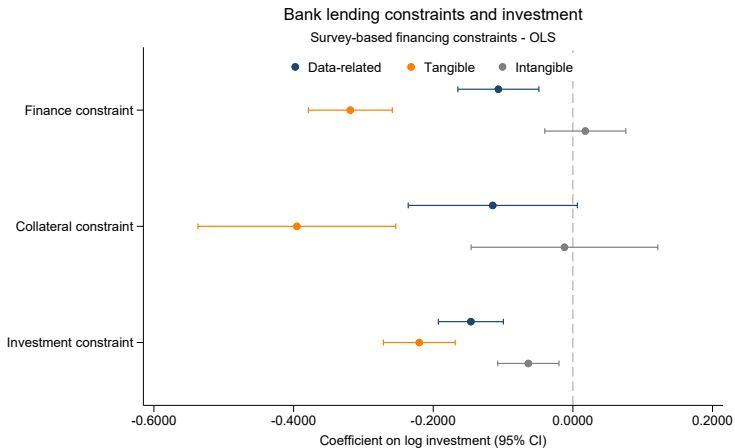
► Manufacturing vs. services

► Regional heterogeneity

Data-related investment is linked to higher productivity

- ▶ Firms investing more in **software and databases** have higher labour productivity and TFP.
- ▶ The relationship is positive on the **intensive margin** and in **zero-inclusive** specifications.
- ▶ IV estimates remain **positive and significant**, providing additional evidence consistent with a causal effect.
 - ▶ No systematic **pre-trends**. ▶ Pre-trends
 - ▶ **Placebo evidence is mixed**. ▶ Placebo
- ▶ The productivity premium is **largest for micro firms** and declines with firm size.

Financing constraints and investment



► Investment / assets

Can cash holdings buffer financing constraints?

	Data-related	Tangible	Intangible
<i>Cash holdings</i>		$N \approx 18,000\text{--}59,000$	
Finance constraint $\times$ Cash	-0.015	0.013	-0.055**
Collateral constraint $\times$ Cash	0.072	0.157**	0.244***
Investment constraint $\times$ Cash	-0.050**	-0.048*	-0.078***
Firm controls	Yes	Yes	Yes
Fixed effects	C $\times$ S $\times$ Y	C $\times$ S $\times$ Y	C $\times$ S $\times$ Y
Cluster	C-S	C-S	C-S

Green: buffering effect. Red: amplification effect. Cash is standardized.

► Implied effects

Financing Constraint IV: Work in progress

Finance Constraint Index		
	Data-related Investment	Tangible Investment
OLS	-0.147*** (0.046)	-0.349*** (0.049)
IV-NPL	-1.295 (1.498)	-4.400** (2.128)
KP rk Wald F	4.17	9.63
Observations	6,665	9,000
Fixed effects	$C \times S \times Y$	$C \times S \times Y$
Cluster	Country	Country

Notes: IV uses bank-side non-performing loan exposure as a credit-supply instrument.

How do financing constraints matter?

- ▶ Survey-based financing constraints are associated with **lower investment**, with the largest and most consistent effects for **tangible investment**.
- ▶ Cash holdings buffer **collateral constraints for tangible and intangible investment**, but not for data-related investment.
 - ▶ Cash interactions
- ▶ Size heterogeneity is **not monotonic**, but micro firms appear particularly exposed to finance and collateral constraints for data-related investment.
 - ▶ Size heterogeneity
- ▶ The bank credit-supply channel remains strongest for **tangible investment**, with little evidence for data-related investment (work in progress).
 - ▶ Bank NPL design
 - ▶ PSM

Conclusion

Conclusion

- ▶ **Result 1:** Data-related investment is associated with **higher labour productivity and TFPR**.
- ▶ **Result 2:** Financing constraints reduce investment, **especially tangible investment**. Bank credit appears less important for data-related investment.
- ▶ **Result 3:** Cash holdings can mitigate **collateral constraints for tangible and intangible investment**, but we find little evidence of a buffering effect for data-related investment.
- ▶ **Policy:**
 - ▷ Bank credit alone is unlikely to unlock data-related investment.
 - ▷ Internal liquidity and alternative finance may be more targeted levers.

Next steps

- ▶ **Extend the data:** broaden the bank match and lengthen the panel to strengthen the credit-supply identification strategy.
- ▶ **Strengthen identification:** develop the regional bank instrument and further validate the compute-based productivity IV.
- ▶ **Understand alternative financing channels:** deepen the evidence on internal liquidity, grants, and other sources of external finance, and identify which constraints they actually relax.

Thank you!

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Appendix

Sample descriptive statistics

Variable	Mean	Median	P75	P99	N
<i>Productivity and investment</i>					
Labour productivity	10.40	10.50	11.10	14.50	81,521
TFPR (ACF)	8.86	8.85	9.65	12.60	57,107
Data-related investment	72,490	2,000	18,565	1,500,000	103,703
Tangible investment	1,000,390	46,869	347,113	19,129,958	103,703
Intangible investment	231,662	10,753	72,164	4,752,041	103,703
AI / Big Data adoption	0.22	0.00	0.00	1.00	65,390
<i>Financing conditions</i>					
ICR < 2	0.17	0.00	0.00	1.00	61,525
Leverage	0.20	0.04	0.19	4.95	84,503
Liquidity	0.16	0.08	0.23	0.86	81,495
Finance constraint	0.09	0.00	0.00	1.00	113,735
Collateral constraint	0.03	0.00	0.00	1.00	43,777
Investment constraint	0.20	0.00	0.00	1.00	117,566
<i>Firm characteristics</i>					
Employees	146	18	75	1,573	119,409
Log total assets	14.3	14.1	15.8	19.7	98,280

Panel structure

Firm panel lengths in the merged EIBIS–Orbis sample

Years observed	Number of firms	Share of firms
1	39,422	62.2%
2	12,248	19.3%
3	5,740	9.1%
4	2,937	4.6%
5	1,428	2.3%
6	806	1.3%
7	418	0.7%
8	224	0.4%
9	161	0.3%
Total	63,384	100.0%

- ▶ Panel length is defined as the number of years in which a firm appears in the merged sample.
- ▶ The dataset is a repeated cross-section with a short firm panel component. [▶ Back](#)

Financing constraint measures

► Survey-based (EIBIS)

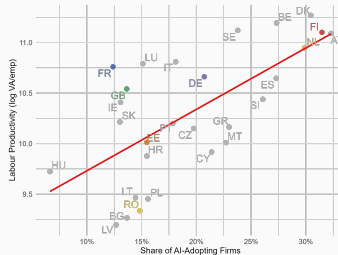
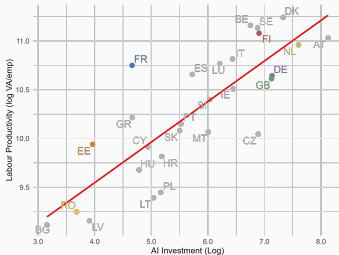
- ▷ **Finance constraint:** 1 if generally financially constrained.
- ▷ **Investment constraint:** 1 if finance is a major obstacle to investment.
- ▷ **Collateral constraint:** 1 if dissatisfied with collateral requirements.
- ▷ **Experience constraint:** 1 if dissatisfied with the terms/outcome of a financing offer.

► Balance-sheet (Orbis)

- ▷ **ICR:** EBITDA / interest expenses; **ICR** < 2: low debt-servicing capacity.
- ▷ **Financial leverage:** financial debt / total assets.

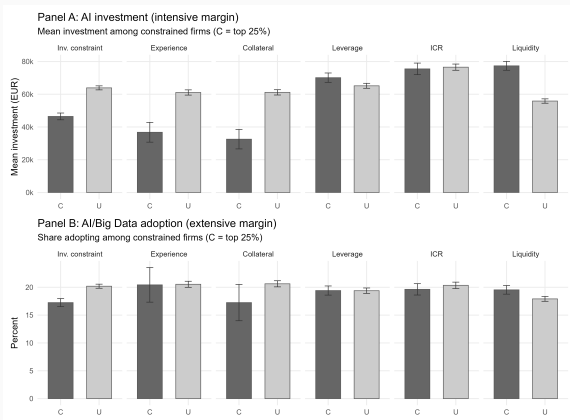
Stylised facts: data-related investment and productivity

- ▶ Countries with higher data-related investment and AI adoption have higher average labour productivity.
- ▶ Strong positive correlation on both the intensive (investment levels) and extensive (adoption share) margins; adoption uneven across Europe.



Financing constraints and data-related investment / adoption

- ▶ Constrained firms (survey-based) invest less in data-related assets and adopt AI/Big Data less frequently.
- ▶ Patterns mixed for balance-sheet measures. [▶ definitions](#)



Identification detail: compute performance-cost measure

$$Z_{i,s,c,t} = Z_t \times Z_{i,s,c}$$

Captures how firms respond differentially to performance-cost changes in computing given pre-existing software/database exposure (Bergeaud 2024).

Component	Variable	Source
Z_t	Cloud compute performance-per-cost measure (time series / US)	Coyle and Hampton (2023)
$Z_{i,s,c}$	Investment share in software (2015)	EIBIS firm-share data

Outcome: TFPR estimated using ACF

	OLS	OLS, IV sample	IV
<i>Investing firms only, log</i>			
Data-related inv.	0.0726*** (0.0090)	0.0737*** (0.0171)	0.0803*** (0.0207)
Observations	28,160	5,386	5,386
First-stage F	—	—	5,609.9
<i>All firms incl. zeros, asinh</i>			
Data-related inv.	0.0136*** (0.0016)	0.0103*** (0.0031)	0.0242*** (0.0058)
Observations	48,261	11,143	11,143
First-stage F	—	—	2,359.3

All specifications: firm controls, $C \times S \times Y$ FE, SE clustered $C \times S$. Instrument: compute performance-per-cost measure $\times$ 2015 data-related investment share. *** $p < 0.01$.

IV validity: tangible-investment placebo

Placebo: replace 2015 data-related investment share with 2015 tangible-investment share.

	Labour productivity	TFPR (ACF)
$\log(I)$	0.161** (0.0803)	0.151 (0.109)
$\operatorname{asinh}(I)$	-0.0300 (0.0519)	0.0144 (0.0450)

- ▶ Placebo estimates are generally **small and statistically insignificant**.
- ▶ The exception is the log labour-productivity specification, which is significant at the 5% level.

Same controls, fixed effects and clustering as the baseline IV. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

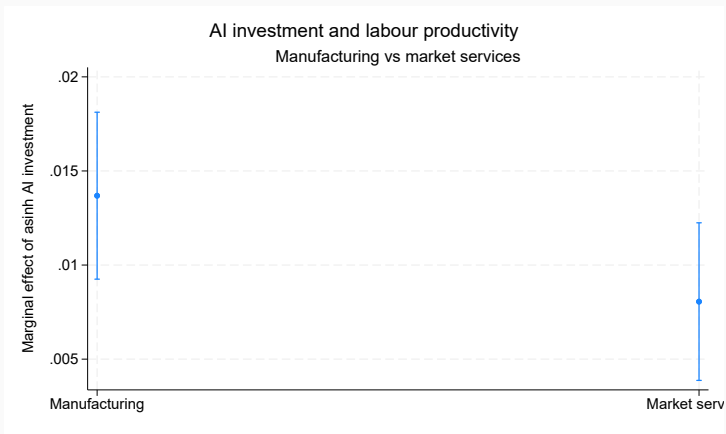
Pre-treatment 2015 data-related investment share $\times$ year coefficients

	Labour productivity	TFPR (ACF)
2010	0.0627 (0.0804)	0.0134 (0.0849)
2011	0.00180 (0.0767)	0.00818 (0.0808)
2012	-0.0239 (0.0633)	0.0337 (0.0649)
2013	-0.00451 (0.0518)	0.0234 (0.0574)
Joint pre-trend p-value	0.865	0.974

- ▶ No individual pre-trend coefficient is statistically significant.
- ▶ Joint tests strongly fail to reject the absence of differential pre-trends.

Standard errors in parentheses. Same controls, FE and clustering as baseline.

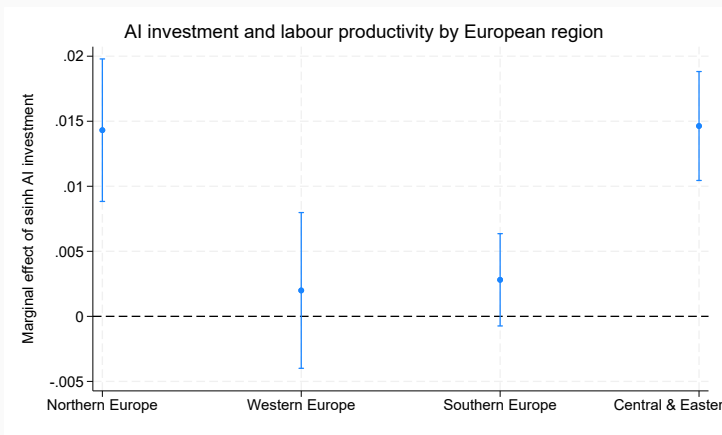
Productivity: manufacturing vs. services



Firm controls; $C \times S \times Y$ FE; SE clustered $C \times S$.

[◀ Back to firm-size heterogeneity](#)

Productivity across regions



Firm controls; $C \times S \times Y$ FE; SE clustered $C \times S$.

[◀ Back to firm-size heterogeneity](#)

Cash holdings: implied constraint effects

	Data-related	Tangible	Intangible
<i>Finance constraint</i>			
Cash: -1 SD	-0.086*	-0.305***	0.083**
Cash: mean	-0.101***	-0.292***	0.028
Cash: +1 SD	-0.116***	-0.279***	-0.028
<i>Collateral constraint</i>			
Cash: -1 SD	-0.190*	-0.505***	-0.190**
Cash: mean	-0.118*	-0.348***	0.054
Cash: +1 SD	-0.046	-0.191**	0.298***
<i>Investment constraint</i>			
Cash: -1 SD	-0.066*	-0.135***	0.050
Cash: mean	-0.116***	-0.183***	-0.028
Cash: +1 SD	-0.166***	-0.231***	-0.106***
Firm controls	Yes	Yes	Yes
Fixed effects	C×S×Y	C×S×Y	C×S×Y
Cluster	C-S	C-S	C-S
Adjusted R^2	0.394–0.399	0.367–0.384	0.390–0.407
Observations	18,064–49,044	24,608–57,741	23,850–59,465

[◀ Back to alternative finance](#)

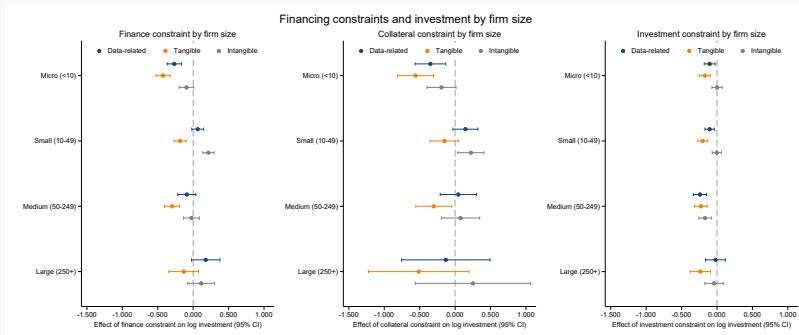
Grant receipt: implied constraint effects

	Data-related	Tangible	Intangible
<i>Finance constraint</i>			
No grants	0.969	-0.035	0.297
Grant recipient	0.004	0.362	0.134
<i>Collateral constraint</i>			
No grants	0.292	0.075	0.295
Grant recipient	0.292	0.075	0.295
<i>Investment constraint</i>			
No grants	-2.474***	-0.639	-1.776**
Grant recipient	-0.057	0.183	0.127
Firm controls	Yes	Yes	Yes
Fixed effects	C×S×Y	C×S×Y	C×S×Y
Cluster	C-S	C-S	C-S
Adjusted R^2	0.247–0.270	0.403–0.406	0.339–0.349
Observations	585–634	839–902	788–857

Grant results should be interpreted cautiously given the small estimation samples.

[◀ Back to alternative finance](#)

Financing constraints and investment by firm size



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Robustness: investment relative to total assets

	Data-related	Tangible	Intangible
Finance constraint	-0.016*** (0.003)	-0.486*** (0.064)	-0.051*** (0.013)
Collateral constraint	-0.012 (0.010)	-0.887*** (0.219)	-0.079** (0.038)
Investment constraint	-0.018*** (0.003)	-0.427*** (0.056)	-0.074*** (0.011)
Firm controls	Yes	Yes	Yes
Fixed effects	C×S×Y	C×S×Y	C×S×Y
Cluster	C-S	C-S	C-S
Adjusted R^2	0.144–0.153	0.156–0.185	0.157–0.174
Observations	30,332–81,658	30,332–81,658	30,332–81,658

Financing constraints are associated with lower investment intensity, with the largest effects for tangible investment. Data-related investment intensity also falls under the finance and investment-constraint measures.

Identification detail: bank credit-supply stress

- ▶ To isolate a **credit-supply channel**, exploit bank-level balance-sheet stress (Brunello et al. 2022).
- ▶ For each firm (i) matched to the main bank ($b(i)$), assign that bank's annual NPL ratio:

$$Z_{it} = \text{NPLratio}_{b(i),t}$$

- ▶ Higher bank NPLs $\Rightarrow$ tighter lending standards, weaker credit supply.
- ▶ Conditional on $c \times s \times t$ FE, NPL variation captures the firm's credit conditions.

Bank credit-supply IV: work in progress

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Coefficient on financial distress, $\mathbf{1}\{\text{ICR} < 2\}$. IV instruments it with bank NPL.

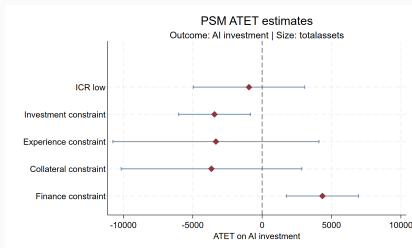
	Outcome: Investment (log)	
	Data-related	Tangible
OLS	-0.190*** (0.046)	-0.539*** (0.054)
IV, using bank NPL exposure	-1.472 (1.093)	-4.727*** (1.585)
KP rk Wald F	11.00	11.32
Observations	6,772	9,159
Fixed effects	C × S × Y	C × S × Y
Cluster	C-S	C-S

First stage moderate ($F \approx 11$). Bank NPL ratio, 2015–2020.

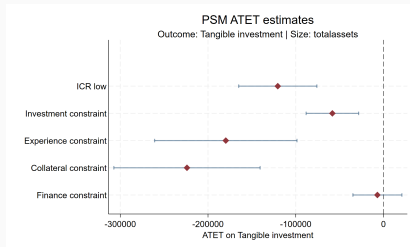
Propensity Score Matching

Constrained versus unconstrained firms; outcome: investment levels including zeros.

Data-related Investment



Tangible Investment



Matched on country, sector, size, year, age.

► Sensitivity & Diagnostics

◄ Back to financing summary

PSM diagnostics: balance and sensitivity

Diagnostics for the matched sample (N = 41,773 matched pairs)

Covariate balance			Rosenbaum sensitivity			
	Before	After	Γ	Effect	95% CI low	95% CI high
Max. SMD	0.241	0.030	1.0	12,212	11,748	12,581
Median SMD	0.036	0.008	1.2	9,344	9,079	16,757
SMD > 0.10	2	0	1.5	6,515	6,229	22,891
			2.0	4,233	4,108	32,437

Note: SMD = Standard mean difference; $\Gamma = 1$ assumes no hidden bias; $\Gamma = 2$ allows matched firms to differ twofold in treatment odds due to unobservables.

- Matching removes observable imbalance: all post-match SMDs are below 0.10.
- The matched-pair estimate remains positive even when hidden bias doubles the odds of treatment within matched pairs.

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